M2 Quantitative Finance

Derivatives Products - Volatility Models

From Heston to Rough Volatility:

Modelling SPX and VIX Options

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1 Introduction

The modelling of implied volatility surfaces remains a central problem in quantitative finance, especially in equity markets where option prices exhibit pronounced deviations from the Black-Scholes framework. In particular, the joint description of SPX options and volatility derivatives such as VIX options requires models capable of capturing both the cross-sectional structure of implied volatility and the temporal dynamics of variance.

This work focuses on stochastic volatility models designed to address these issues. Starting from the classical Heston model, which provides a tractable affine framework with stochastic variance and leverage effects, we then consider its rough extension, the Rough Heston model, which incorporates fractional dynamics in order to reproduce the empirically observed roughness of volatility and the steep SPX smile behavior (Gatheral et al., 2018; El Euch and Rosenbaum, 2019). We also analyze the Lifted Heston model as a tractable finite-dimensional approximation of Rough Heston, and finally present a joint SPX-VIX calibration

The objective of the study is twofold. On the one hand, we aim to analyze the theoretical structure of these models, with particular emphasis on their variance dynamics, transform-based pricing methods, and the role of forward variance in linking SPX and VIX markets. On the other hand, we seek to assess their practical relevance by comparing their ability to reproduce market data, both in terms of SPX implied volatility surfaces and VIX option prices.

2 Heston and Rough Heston Models

The Heston model provides a natural starting point for the analysis. It is the classical benchmark within affine stochastic volatility models: it departs from the Black–Scholes framework by allowing the variance to evolve randomly over time, while preserving a structure that leads to transform-based pricing methods (Heston, 1993). The Rough Heston model is then introduced as a fractional extension of the same idea, obtained by replacing the classical mean-reverting CIR variance dynamics with a rough Volterra-type specification. This extension allows the model to capture the lower temporal regularity observed in volatility data and the more pronounced SPX smile curvature visible in equity index options (El Euch and Rosenbaum, 2019; Pulido, 2026).

2.1 The Heston Model

The Heston model is one of the canonical stochastic volatility models. Under the risk-neutral measure $\mathbb{Q}$, the asset price process $(S_t)_{t\geq 0}$ and its instantaneous variance process $(V_t)_{t\geq 0}$ satisfy

$$dS_t = rS_t\,dt + \sqrt{V_t}\,S_t\,dB_t, \tag{1}$$

$$dV_t = \lambda(\theta - V_t)\,dt + \nu\sqrt{V_t}\,dW_t, \tag{2}$$

with

$$d\langle B, W\rangle_t = \rho\,dt. \tag{3}$$

Here r is the short rate, assumed constant, and (B, W) is a two-dimensional Brownian motion under $\mathbb{Q}$ (Heston, 1993). Equivalently, writing

$$B_t = \rho W_t + \sqrt{1-\rho^2}\, W_t^{\perp},$$

with $W^{\perp}$ independent of W, the stock-price equation becomes

$$dS_t = rS_t\, dt + \sqrt{V_t}\, S_t\big(\rho\, dW_t + \sqrt{1-\rho^2}\, dW_t^{\perp}\big). \tag{4}$$

This representation isolates the leverage channel: part of the return shock is generated by the same Brownian motion that drives variance. The second equation is a Cox-Ingersoll-Ross type dynamics. In particular, the diffusion coefficient is proportional to $\sqrt{V_t}$, which is the natural mechanism ensuring that volatility shocks vanish as the variance approaches zero. In the lecture notes, the condition $\lambda\theta \geq 0$ is highlighted as ensuring the existence of a unique non-negative solution (Pulido, 2026).

2.1.1 Conditional moments and forward variance

A key property of the Heston variance process is mean reversion, which leads to an explicit expression for the conditional expectation of future variance. In particular, under the risk-neutral measure,

$$\mathbb{E}^{\mathbb{Q}}[V_u \mid \mathcal{F}_t] = \theta + (V_t - \theta)e^{-\lambda(u-t)}. \tag{5}$$

This formula provides the forward variance curve in closed form and is particularly important for pricing variance swaps and VIX-related derivatives, that depend directly on expected future variance levels.

2.1.2 Role of the parameters

The model is determined by the parameter vector

$$\Theta = (V_0, \lambda, \theta, \nu, \rho).$$

The parameter V_0 fixes the initial level of spot variance and therefore has a direct influence on short-dated option prices. The parameter θ is the long-run level toward which variance reverts. The mean-reversion speed λ controls how rapidly the variance process forgets its current level. When λ is large, the process reverts quickly and the forward variance curve flattens rapidly toward θ. When λ is small, shocks to variance persist over longer horizons. The parameter ν is the volatility of volatility. It determines the magnitude of random fluctuations in V_t and therefore controls the dispersion of future integrated variance. In option prices, this mainly affects smile convexity and tail thickness. In Heston (1993), the volatility-of-volatility parameter is shown to affect primarily the kurtosis of the risk-neutral return distribution: an increase in ν amplifies fluctuations in the variance process, there by generating heavier tails and, in turn, more pronounced curvature in option prices and implied volatility smiles. The correlation parameter ρ plays a different role. It controls the dependence between return shocks and variance shocks. This parameter is crucial because it is the main source of asymmetry in the risk-neutral distribution of returns. In equity markets, ρ is typically negative, reflecting the leverage effect: negative returns tend to be accompanied by upward revisions in volatility (Heston, 1993; Pulido, 2026). Consequently, negative ρ generates the negative skew observed in index option markets. This separation

is conceptually important: in Heston, ν mostly controls curvature, whereas ρ controls skew.

2.1.3 Affine structure and characteristic function

The pair $(\log S_t, V_t)$ is an affine process, so the characteristic function of the log-price has an exponential-affine form in the initial variance (Heston, 1993). In particular,

$$\mathbb{E}\big[e^{uX_T} \mid \mathcal{F}_0\big] = \exp\big(\phi(T;u) + \psi(T;u)V_0 + uX_0\big), \tag{6}$$

where $\phi(\cdot;u)$ and $\psi(\cdot;u)$ solve a Riccati system. In particular, ψ satisfies a scalar nonlinear Riccati differential equation, and the pair (ϕ, ψ) fully determines the law of $\log S_T$.

The same affine structure also underlies Heston's original semi-closed pricing formula for European calls:

$$C(S_t, V_t, t) = S_t P_1 - K e^{-r(T-t)} P_2, \tag{7}$$

where

$$P_j(x, v, t; \log K) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re\left(\frac{e^{-i\varphi \log K} f_j(x, v, t; \varphi)}{i\varphi}\right) d\varphi, \qquad j = 1, 2, \tag{8}$$

and

$$f_j(x, v, t; \varphi) = \exp\big(C_j(T-t;\varphi) + D_j(T-t;\varphi)v + i\varphi x\big). \tag{9}$$

Here $C_j(\cdot;\varphi)$ and $D_j(\cdot;\varphi)$ are obtained by evaluating the affine Riccati solution at the complex arguments used in the pricing formula.

The Riccati system reads

$$\phi'(t;u) = \lambda\theta\, \psi(t;u), \qquad \phi(0;u) = 0, \tag{10}$$

$$\psi'(t;u) = \frac{1}{2}(u^2 - u) + (u\rho\nu - \lambda)\psi(t;u) + \frac{\nu^2}{2}\psi(t;u)^2, \qquad \psi(0;u) = 0. \tag{11}$$

The practical advantage is that vanilla option prices can be computed by one-dimensional numerical integration rather than simulation. An analogous representation holds for European puts.

2.1.4 Pricing SPX options

For SPX options, the pricing problem is therefore reduced to the computation of the risk-neutral law of $\log S_T$. In practice, one fixes (K, T) and evaluates the model price either through Heston's semi-closed formula (7)-(8) or through a generic Fourier inversion formula based on the characteristic function of X_T. If $\hat{q}$ denotes the characteristic function of X_T and f the payoff expressed as a function of log-price, then the option price can be written in the damped Fourier form

$$C_0 = e^{-rT} \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(u + i\omega)\hat{q}(u + i\omega)\, du, \tag{12}$$

for a suitable damping parameter ω (Pulido, 2026). The advantage of this representation is computational. Calibration requires evaluating the model repeatedly on many strikes and maturities; transform methods are therefore much more efficient than Monte Carlo for vanilla SPX options.

2.1.5 VIX and VIX options

The link with the VIX market follows directly from the forward variance curve. For the S&P 500, the squared VIX is interpreted as the risk-neutral average of future variance over a fixed horizon Δ (typically 30 days),

$$VIX_t^2 = \frac{1}{\Delta}\int_t^{t+\Delta} \xi_t(u)\,du. \tag{13}$$

Using the explicit Heston forward variance formula, this becomes

$$VIX_t^2 = \theta + (V_t - \theta)\frac{1-e^{-\lambda\Delta}}{\lambda\Delta}. \tag{14}$$

Hence, the squared VIX is an explicit affine function of the current variance V_t, so that pricing VIX options reduces to evaluating expectations of functions of V_T. In particular, a European call with maturity T and strike K is given by

$$C_t^{VIX} = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}\big[(VIX_T - K)^+ \mid \mathcal{F}_t\big]. \tag{15}$$

Unlike SPX options, no semi-closed Fourier formula is available in this framework, so pricing typically relies on numerical methods such as Monte Carlo simulation of the variance process. For market comparison, it is useful to introduce the VIX futures price

$$F_t^{VIX}(T) := \mathbb{E}^{\mathbb{Q}}[VIX_T \mid \mathcal{F}_t],$$

which provides the forward level of the VIX at option maturity and is the natural input for Black implied-volatility quoting.

2.2 The Rough Heston Model

The Rough Heston model extends the classical Heston framework by replacing the Markovian CIR variance dynamics with a fractional Volterra equation, introducing memory effects and non-Markovian behavior. Under the risk-neutral measure $\mathbb{Q}$, the stock price and variance processes satisfy

$$dS_t = rS_t\,dt + \sqrt{V_t}\,S_t\,dB_t, \tag{16}$$

$$V_t = V_0 + \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\lambda(\theta - V_s)\,ds + \frac{1}{\Gamma(\alpha)}\int_0^t (t-s)^{\alpha-1}\nu\sqrt{V_s}\,dW_s, \tag{17}$$

with

$$d\langle B, W\rangle_t = \rho\,dt, \qquad \alpha \in \left(\frac{1}{2}, 1\right). \tag{18}$$

The key difference with respect to the classical Heston model lies in the fractional kernel

$$K(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)},$$

which makes the variance depend on its whole past trajectory through power-law memory. The param-

eter α controls the path regularity. Defining

$$H = \alpha - \frac{1}{2},$$

smaller values of H correspond to rougher variance paths, which are highly irregular and display the rough behavior observed empirically in volatility time series (Gatheral et al., 2018; Pulido, 2026). When $\alpha = 1$, the model reduces to the standard Heston variance equation.

2.2.1 Role of the parameters

The parameter vector is

$$\Theta = (V_0, \lambda, \theta, \nu, \rho, \alpha).$$

The parameters $(V_0, \lambda, \theta, \nu, \rho)$ retain the same economic interpretation as in the standard Heston model, but their effect is now mediated by the Volterra kernel. The initial variance V_0 fixes the short-end level of volatility. The parameter θ is the long-run mean-reversion level, while λ controls the strength of the pull toward that level. In contrast with the Markovian Heston case, however, this mean reversion is no longer local in time: the contribution of the term $\lambda(\theta - V_s)$ at past date s is weighted by $(t-s)^{\alpha-1}$ when determining V_t. The parameter ν is the volatility of volatility and controls the amplitude of random fluctuations in the variance process. As in Heston, larger values of ν increase the variability of future variance and strengthen stochastic-volatility effects in option prices. The parameter ρ determines the leverage effect. When $\rho < 0$, downward shocks to the underlying are associated with upward shocks to variance, which generates the negative skew characteristic of equity-index option markets (El Euch and Rosenbaum, 2019; Pulido, 2026). Finally, α is the roughness parameter: it controls the temporal regularity of the variance trajectories and therefore the short-maturity behavior of the implied-volatility skew.

2.2.2 Affine-type structure and characteristic function

Despite losing finite-dimensional Markovianity, the Rough Heston model remains tractable because the log-price characteristic function preserves an exponential-affine structure (El Euch and Rosenbaum, 2019). Let

$$X_t := \log\left(\frac{S_t}{S_0}\right) - rt.$$

Then the characteristic function can be written as

$$\mathbb{E}\big[e^{zX_T}\big] = \exp\Big(\theta\lambda I^1 h(z,T) + V_0 I^{1-\alpha} h(z,T)\Big), \tag{19}$$

where the function h solves the fractional Riccati equation

$$D^\alpha h(z,t) = \frac{1}{2}(z^2 - z) + (z\rho\nu - \lambda)h(z,t) + \frac{\nu^2}{2}h(z,t)^2. \tag{20}$$

The key difference with respect to the classical Heston model is that the standard Riccati ordinary differential equation is replaced by a fractional Riccati equation. As a result, transform-based pricing remains available, although at a significantly higher numerical cost.

2.2.3 Pricing SPX options

SPX option pricing in the Rough Heston model follows the same Fourier inversion framework used in the classical Heston case. Once the characteristic function of X_T has been obtained from the fractional Riccati equation, the option price is computed as

$$C_0 = e^{-rT} \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(u + i\omega)\hat{q}(u + i\omega)\, du, \tag{21}$$

for a suitable damping parameter ω (El Euch and Rosenbaum, 2019; Pulido, 2026). The conceptual pricing structure is therefore unchanged with respect to the classical Heston model: prices are still obtained by Fourier inversion of the log-price characteristic function. The main difference is computational, since the characteristic function is now determined by a fractional Riccati equation rather than a standard Riccati ODE. In practice, the fractional Riccati equation is rewritten in equivalent Volterra integral form. The resulting nonlinear Volterra equation is then solved numerically using a fractional Adams predictor-corrector scheme (El Euch and Rosenbaum, 2019). As a result, the main computational cost comes from repeatedly solving this equation during calibration.

2.2.4 VIX and VIX options

In the Rough Heston framework, the key object linking SPX and VIX products is the forward variance curve

$$\xi_t(u) = \mathbb{E}^{\mathbb{Q}}[V_u \mid \mathcal{F}_t],$$

which summarizes the information carried by the past variance trajectory (El Euch and Rosenbaum, 2018). Unlike the classical Heston model, the future VIX is not a simple function of the scalar variance level V_T, but depends on the whole forward variance curve beyond maturity. A VIX call with maturity T and strike K is therefore priced under the standard risk-neutral expectation

$$C_t^{VIX} = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}\big[(VIX_T - K)^+ \mid \mathcal{F}_t\big]. \tag{22}$$

Since the payoff depends on an infinite-dimensional state variable, no semi-closed pricing formula is available as in the SPX case. In practice, VIX options are therefore priced numerically, typically through Monte Carlo simulation of the rough variance dynamics, and then converted into Black implied volatilities using the corresponding VIX futures price as forward input.

2.2.5 The Lifted Rough Heston Model

The Lifted Rough Heston model provides a finite-dimensional approximation of Rough Heston by replacing the fractional kernel

$$K(t) = \frac{t^{\alpha-1}}{\Gamma(\alpha)}$$

with a finite sum of exponentials

$$K^n(t) = \sum_{i=1}^{n} c_i e^{-x_i t},$$

where the coefficients are chosen so that K^n approximates the rough kernel as n increases (El Euch and Rosenbaum, 2018).

Under this representation, the variance process becomes a sum of n mean-reverting factors,

$$V_t^n = \sum_{i=1}^{n} U_t^{(i)}, \quad where \quad dU_t^{(i)} = -x_i U_t^{(i)}\, dt + c_i \lambda(\theta - V_t^n)\, dt + c_i \nu \sqrt{V_t^n}\, dW_t. \tag{23}$$

so that the model is Markovian in the extended state vector $(S_t, U_t^{(1)}, \ldots, U_t^{(n)})$. The main advantage of the lifted formulation is that it preserves the main rough-volatility effects while remaining computationally tractable and easier to calibrate than the fully rough specification (El Euch and Rosenbaum, 2018).

2.3 Calibration methodology

For all the models, calibration was performed by minimising a weighted mean squared error on implied volatilities, with weights inversely proportional to the relative bid-ask spread (more liquid quotes receive higher weight):

$$\mathcal{L}(\Theta) = \sum_i w_i \big(\sigma_i^{\text{model}}(\Theta) - \sigma_i^{\text{market}}\big)^2. \tag{24}$$

The numerical optimizer was L-BFGS-B with multi-start initialization. For the Heston model, a Feller penalty was added to the objective to discourage parameter configurations where the condition $2\lambda\theta \geq \nu^2$ (which ensures that V_t remains strictly positive) is violated. For the rough Heston model, H was fixed on a discrete grid and the five remaining parameters $(\lambda, \theta, \nu, \rho, V_0)$ were jointly optimized for each value of H, the best H was then selected a posteriori. A full joint optimisation over all six parameters simultaneously is impractical with gradient-based methods due to the non-smooth dependence of the Adams solver on H.

2.4 Heston Model - Calibration Results

After a two-stage multi-start calibration (coarse grid with 6 starts, then fine refinement with 3 starts), the Heston model converged in approximately 30 seconds. The calibrated Heston parameters obtained from the SPX implied volatility fit are:

$$\lambda = 4.6782, \qquad \theta = 0.0448, \qquad \xi = 0.6457, \qquad \rho = -0.7792, \qquad v_0 = 0.0168.$$

The calibrated parameters are economically plausible: the high λ implies fast mean reversion, the strongly negative ρ captures the SPX leverage effect, and the large ξ is consistent with the observed smile steepness. The Feller condition $2\lambda\theta = 0.4189 > \xi^2 = 0.4169$ is satisfied, guaranteeing that the variance process remains strictly positive almost surely. The fit quality on SPX implied volatilities is summarized using three standard metrics: the root mean squared error (RMSE), the mean absolute error (MAE) and the weighted mean squared error. We obtain: RMSE = 2.31 vol pts, MAE = 1.68 vol pts and Weighted MSE = 0.0003. Overall, these values indicate a satisfactory calibration performance, with the weighted loss confirming that the model captures the most relevant regions of the implied-volatility surface with reasonable accuracy.

2.4.1 Heston Model - Comparison between market and model implied volatilities for SPX

Overall, the graphical comparison confirms that the Heston model provides a reasonable fit to the SPX implied-volatility surface. The model successfully captures the general level of implied volatility and the overall negative slope of the smile across maturities. The main discrepancy is primarily visible in the wings, where the market implied-volatility smile displays a more pronounced curvature than the smoother profile generated by the standard Heston volatility dynamics. This effect is particularly noticeable at low and high strikes, where the market smile bends more sharply than the model-implied curve. However, fit quality decreases as maturities increase.

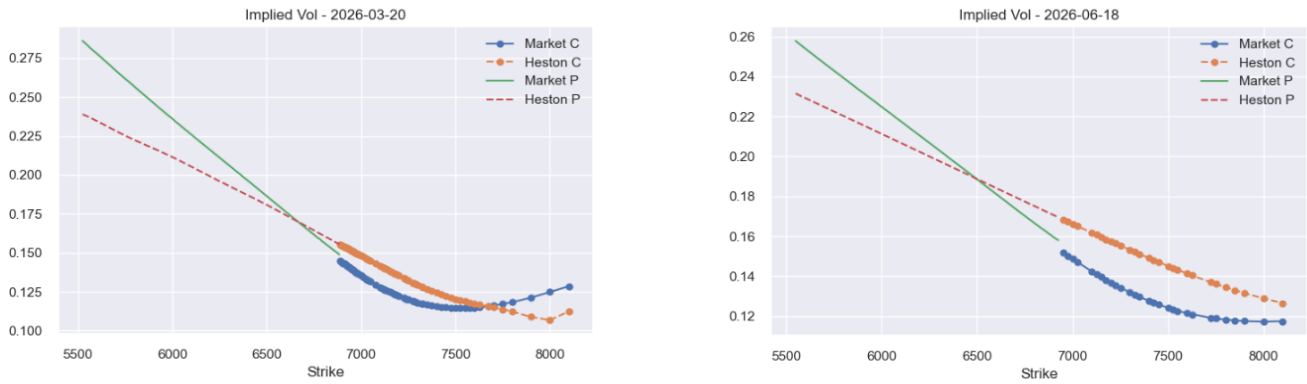

Figure 1: Comparison between market SPX implied volatilities and Heston model-implied volatilities for selected maturities. (Results are available for the full set of maturities in the accompanying Jupyter notebook)

2.4.2 Heston Model - Model VIX options versus market VIX options

The Heston-implied VIX smiles reproduce the overall upward slope across strikes, preserving the general increasing shape observed in market quotes. However, a substantial level mismatch remains across both maturities. In particular, the model systematically underestimates the call while significantly overestimating the put-side implied-volatility level. This discrepancy suggests that an SPX-only calibration is not sufficient to accurately reproduce VIX option prices. The out-of-sample fit confirms this limitation quantitatively, with RMSE = 63.01 vol pts and MAE = 43.59 vol pts.

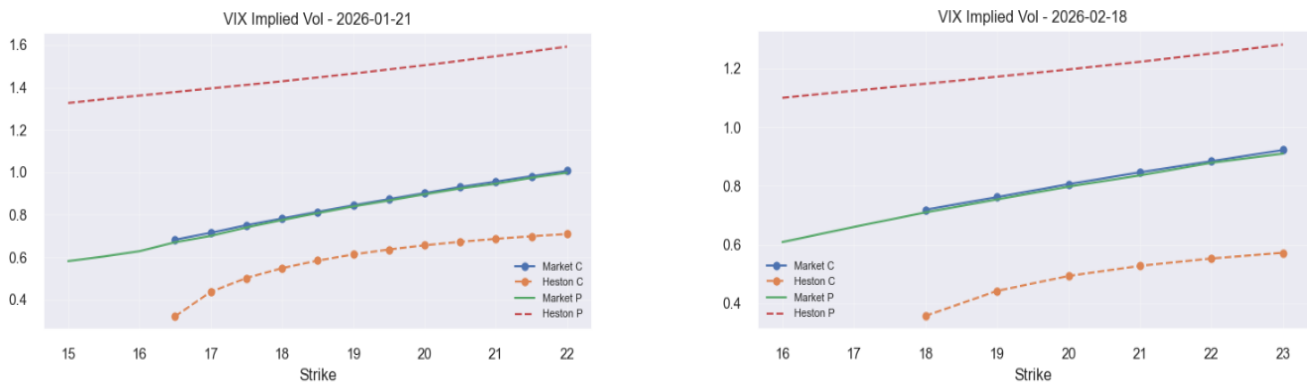

Figure 2: Comparison between market VIX implied volatilities and model-implied VIX implied volatilities obtained from SPX-calibrated Heston for selected maturities. (Results are available for the full set of maturities in the accompanying Jupyter notebook)

2.4.3 Heston Model - Fit quality

The Heston model provides a solid benchmark fit for the SPX implied-volatility surface, successfully capturing the overall smile level and the general negative skew structure. The skew term structure is reasonably captured at short and intermediate maturities, while the long-end comparison is less informative because fewer weighted quotes remain after filtering. However, the fit quality deteriorates in the wings, where the market smile exhibits a stronger curvature than the smoother profile generated by the model. The out-of-sample VIX comparison confirms this limitation, as the model reproduces the general upward slope across strikes but exhibits a substantial level mismatch relative to market quotes. Overall, the model remains a reasonable parsimonious baseline, although its structural flexibility is limited when compared with more advanced volatility models.

2.5 Rough Heston Model - Calibration Results

The Rough Heston calibration is performed on the same SPX option dataset as the Heston benchmark, using a slightly tighter moneyness filter to improve numerical stability and focus the fit on the central smile region. To avoid numerical instabilities induced by the fractional Adams solver, the roughness level is selected through a two-stage grid search, while the remaining parameters are optimized conditionally for each fixed grid value.

The optimal calibrated parameters are:

$$H = 0.1850, \quad \lambda = 0.8068, \quad \theta = 0.0847, \quad \nu = 0.5417, \quad \rho = -0.7092, \quad V_0 = 0.0214.$$

The estimated roughness level remains consistent with rough-volatility dynamics and supports the stronger smile curvature observed in the SPX surface. Relative to the classical Heston benchmark, the model delivers a significantly lower calibration error, with RMSE $= 0.84$ vol pts, MAE $= 0.53$ vol pts, and Weighted MSE $= 0.000033$, corresponding to a substantial improvement in fit quality.

2.5.1 Rough Heston Model - Comparison between market and model implied volatilities for SPX

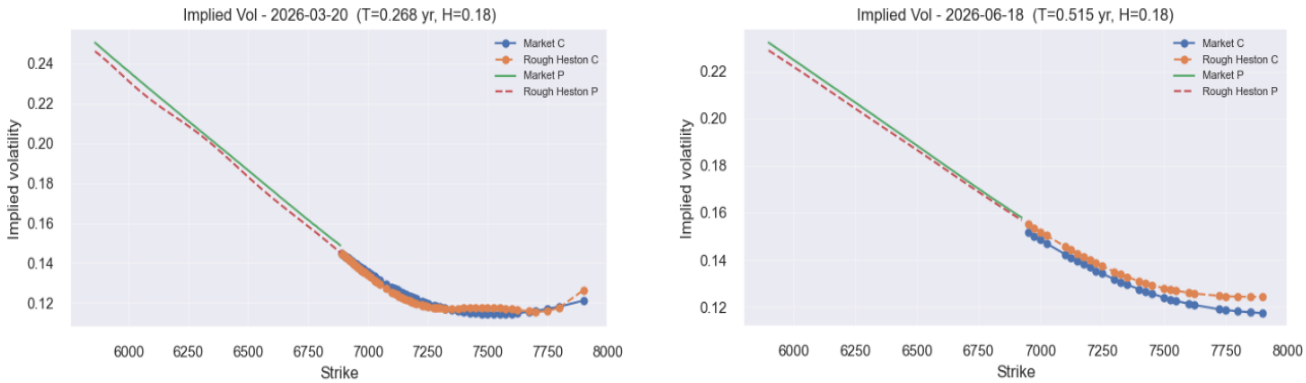

Figure 3: Comparison between market SPX implied volatilities and Rough Heston model- implied volatilities for selected maturities. (Results are available for the full set of maturities in the accompanying Jupyter notebook)

The Rough Heston model reproduces the SPX implied-volatility smile with visibly higher accuracy across both selected maturities. The model-implied curves closely overlap market quotes across the central strike region, capturing both the overall smile level and its curvature with high precision. Only minor residual discrepancies remain in the far right wing, where the model slightly overestimates implied volatility at high strikes.

2.5.2 Rough Heston Model - Model VIX options versus market VIX options

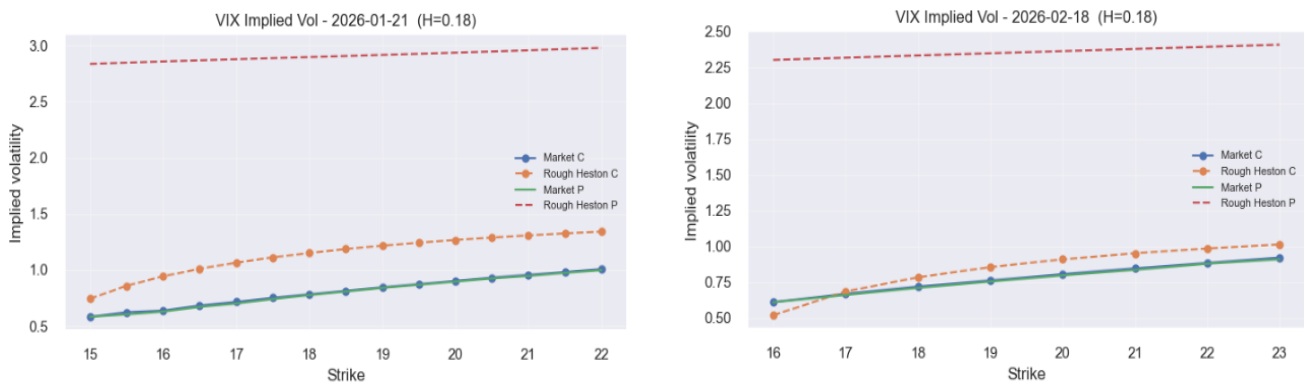

Figure 4: Comparison between market VIX implied volatilities and model-implied VIX implied volatilities obtained from SPX-calibrated Rough Heston for selected maturities. (Results are available for the full set of maturities in the accompanying Jupyter notebook)

Despite the clear improvement on the SPX surface, the Rough Heston model provides a less satisfactory VIX out-of-sample fit, with implied volatility levels that remain systematically too high across strikes. This limitation is also reflected in the out-of-sample error metrics, with RMSE = 151.37 vol pts and MAE = 105.65 vol pts, confirming that the SPX-only calibration does not fully capture the dynamics embedded in VIX option prices.

2.5.3 Rough Heston Model - Fit quality

The Rough Heston model significantly improves the SPX implied-volatility fit, delivering a much more accurate reproduction of both the smile level and its curvature across the selected maturities. The skew term structure is also better captured at short and intermediate maturities, while the long-end comparison remains less informative because fewer filtered quotes are available. The calibrated roughness level $H = 0.185$ remains consistent with rough-volatility dynamics and supports the improved fit quality observed on the SPX surface. However, the VIX out-of-sample fit remains unsatisfactory, as the model still overestimates implied-volatility levels, particularly on the put side. This limitation is reflected in the large pricing errors (RMSE = 151.37 vol pts and MAE = 105.65 vol pts), which remain worse than those obtained under the classical Heston benchmark (RMSE = 63.01 vol pts, MAE = 43.59 vol pts), although the difference is mainly quantitative rather than structural, since neither model is able to reproduce VIX smiles accurately under an SPX-only calibration. Overall, despite the superior SPX fit, an SPX-only Rough Heston calibration remains insufficient to accurately reproduce VIX option prices and does not materially improve long-maturity consistency.

3 Models' Results Comparison

The Rough Heston model clearly outperforms the classical Heston benchmark in terms of SPX calibration quality. Across maturities, the SPX RMSE is consistently lower, with an aggregate RMSE of 0.84 vol pts versus 2.31 vol pts for Heston. The improvement is particularly visible at short and intermediate maturities, where the rough dynamics better reproduce smile curvature and the skew term structure. At longer maturities, however, the comparison on skew term structure becomes less informative because fewer strike quotes remain available after filtering, a feature common to both models and mainly driven by lower market liquidity. As a result, the long-end skew reproduction cannot be assessed with the same level of confidence. This superior SPX fit does not translate into better VIX consistency. The Rough Heston model produces a significantly worse out-of-sample VIX fit, with RMSE = 151.37 vol pts and MAE = 105.65 vol pts, compared with RMSE = 63.01 vol pts and MAE = 43.59 vol pts under the classical Heston benchmark. This suggests that the improved SPX smile fit comes at the cost of a less stable forward variance representation. From a computational perspective, the Heston model is substantially faster (approximately 30 s), since pricing relies on a standard Riccati equation combined with Fourier inversion, whereas Rough Heston requires repeatedly solving a fractional Riccati equation through the Adams predictor-corrector scheme. This makes calibration significantly more expensive and numerically less stable, leading to a total runtime of approximately 95 s. Overall, the comparison highlights a clear trade-off: Rough Heston provides a much better short-end SPX fit and a more realistic skew term structure, while classical Heston remains faster and more robust in terms of VIX consistency.

4 Simulation and Regularity Studies

4.1 Heston Model

Using the calibrated Heston parameters, we simulate one-year trajectories of both the asset price and the variance process under the full-truncation Euler scheme. Figure 5 reports representative simulated price and variance paths.

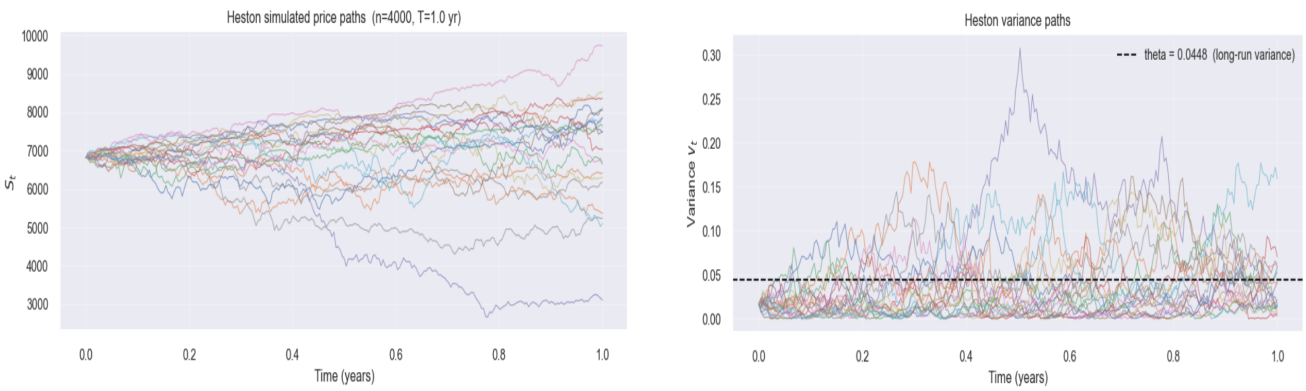

Figure 5: Simulated Heston price and variance paths.

The variance paths clearly fluctuate around the long-run level $\theta = 0.0448$, consistently with the mean-reverting structure of the CIR-type variance dynamics. To study path regularity, we estimate the scaling law of second moments of increments across increasingly fine discretization levels. The analysis is

performed over multiple lag scales ranging from coarse to fine grids, corresponding to $h, 2h, 4h, \ldots, 64h$:

$$\mathbb{E}\left[|X_{t+h} - X_t|^2\right] \sim h^{2H}.$$

A log-log regression over multiple lag sizes yields to:

$$\widehat{H}_{\log S} = 0.5041, \qquad \widehat{H}_v = 0.4859$$

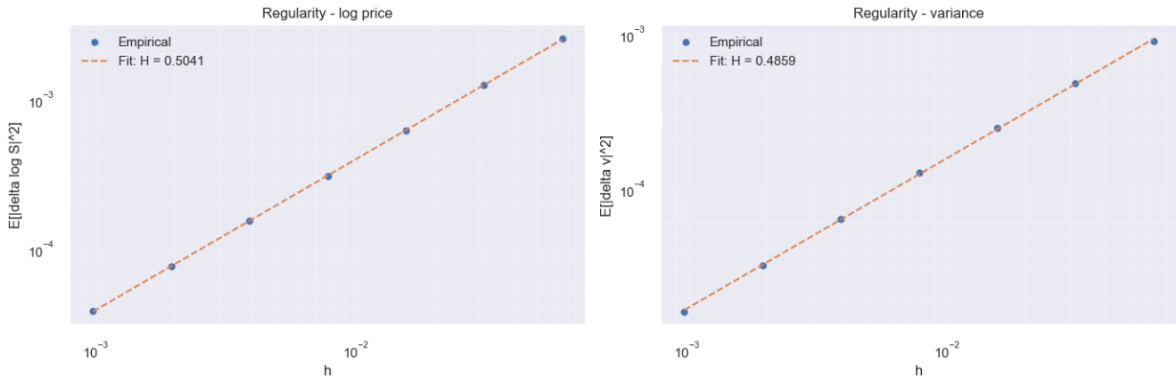

Figure 6: Log-log increment scaling for simulated Heston log-price and variance paths, with $\widehat{H} \approx 0.5$.

These estimates are fully consistent with the theoretical $H = 0.5$ regularity of standard diffusions. Moreover, the log-log scaling remains remarkably stable across progressively finer lag scales, indicating that the numerical estimate of H remains robust as the effective discretization becomes finer.

4.2 Rough Heston Model

Using the calibrated Rough Heston parameters, we simulate one-year trajectories of the variance process and study its regularity across multiple time discretizations, from coarse to fine grids. The analysis is based on the scaling relation

$$\mathbb{E}\left[|X_{t+h} - X_t|\right] \sim h^H,$$

evaluated over the discretization sequence $N = 64, 128, 256, 512, 1024$. A log-log regression of the empirical increment moments yields $\widehat{H} = 0.2060$ and $H_{\text{theoretical}} = 0.1850$.

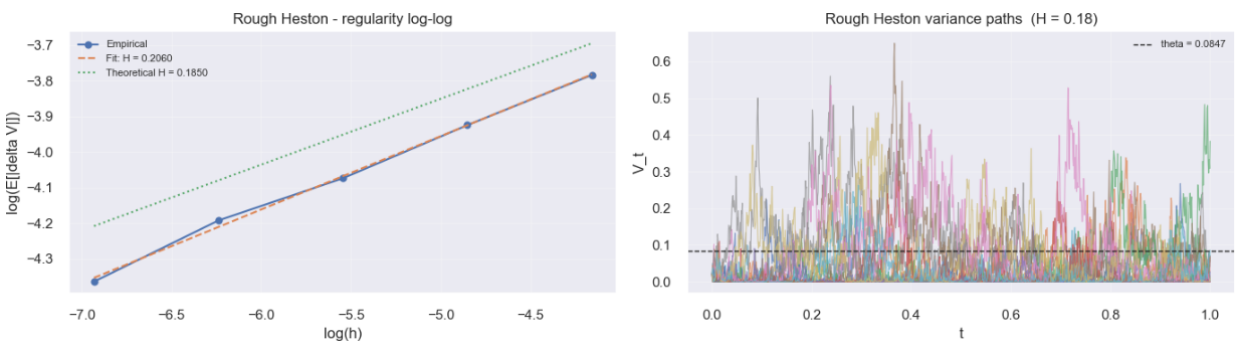

Figure 7: Rough Heston regularity study: log-log scaling of variance increments and representative simulated variance paths.

The estimated exponent remains close to the theoretical value as the discretization becomes finer, indicating that the scaling behavior is numerically stable across grids. The variance trajectories are markedly more irregular than in the classical Heston case, and the estimate $\widehat{H} \approx 0.2060$ is fully consistent with the rough-volatility nature of the model.

5 Lifted Rough Heston Model Improvements

5.1 Calibration

As an additional benchmark to improve the overall comparison between Heston and Rough Heston, we also include the Lifted Heston model. This model approximates the Rough Heston dynamics with a finite number of factors, making it easier to simulate and calibrate while still preserving rough-volatility features. The calibration is performed on the same filtered SPX implied volatility dataset used for the Rough Heston model, using a two-stage grid search over the parameter H, which controls the path roughness. For each fixed H, the remaining parameters $(\lambda, \theta, \nu, \rho, V_0)$ are estimated by minimizing the same liquidity-weighted least-squares criterion on implied volatility errors.

The optimal calibrated parameters are $H = 0.1100$, $\lambda = 0.4830$, $\theta = 0.2240$, $\nu = 0.7460$, $\rho = -0.7160$, and $V_0 = 0.0146$. The low estimated value of H confirms the presence of rough-volatility effects, while the finite number of factors makes the model computationally easier than the fully rough specification. The SPX implied-volatility fit is also very accurate, with RMSE = 0.99 vol pts, MAE = 0.64 vol pts, and weighted MSE = 0.000049.

5.2 Lifted Heston Model - Graphical comparison between market and model implied volatilities for SPX

The Lifted Heston model reproduces the SPX implied-volatility smile very accurately across both selected maturities. The market skew slope and the central smile curvature are closely matched, with only small deviations in the far right wing. The fit remains stable across short and intermediate maturities, while the long-end comparison is less informative because fewer filtered quotes are available, showing that the finite-factor approximation preserves most of the flexibility needed to capture the SPX surface.

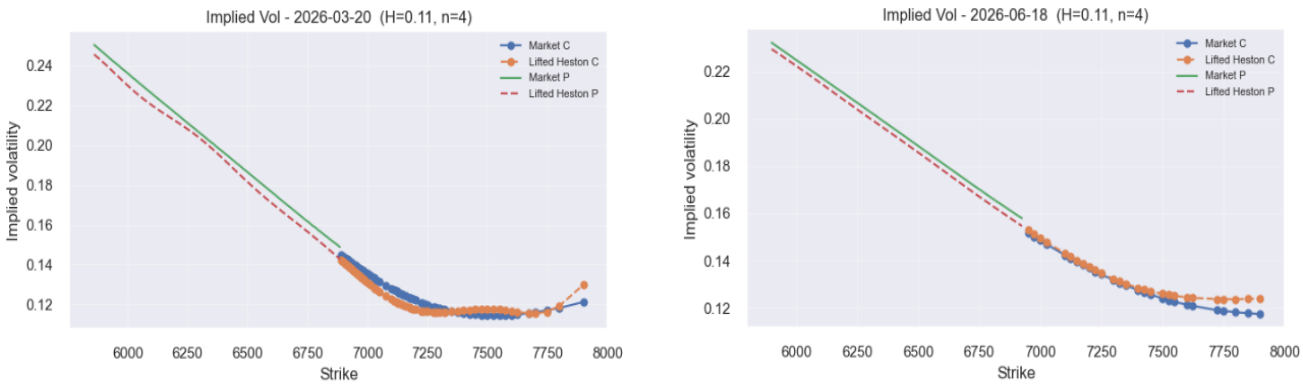

Figure 8: Comparison between market SPX implied volatilities and Lifted Heston model-implied volatilities for selected maturities. (Results are available for the full set of maturities in the accompanying Jupyter notebook)

5.3 Lifted Heston Model - Graphical comparison for VIX options

The Lifted Heston model provides a weak out-of-sample fit for VIX implied volatilities, despite its strong SPX calibration performance. The model systematically underestimates the call-side VIX and generates a put-side level that remains far above market quotes. However, the VIX fit improves relative to the Rough Heston, as reflected in the lower error metrics (RMSE = 133.43 vol pts and MAE = 89.45 vol pts, versus RMSE = 151.37 vol pts and MAE = 105.65 vol pts under Rough Heston). This indicates a partial recovery in SPX-VIX consistency, although the overall mismatch remains substantial under an SPX-only calibration.

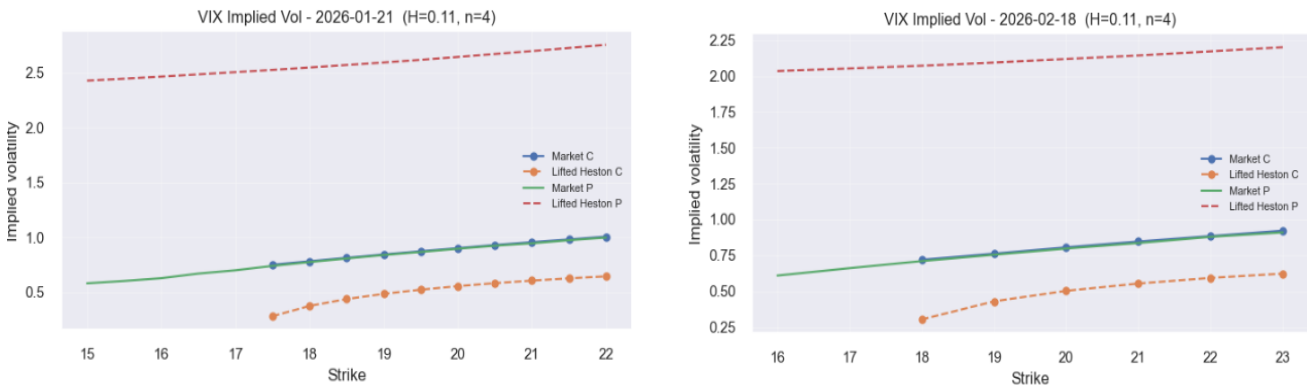

Figure 9: Comparison between market SPX implied volatilities and Lifted Heston model-implied volatilities for selected maturities. (Results are available for the full set of maturities in the accompanying Jupyter notebook)

5.4 Simulation and Regularity

Using the calibrated Lifted Heston parameters, we simulate one-year trajectories of the variance process and study the scaling of increments across progressively finer discretization levels. The analysis is based on the scaling relation

$$\mathbb{E}\left[|X_{t+h} - X_t|\right] \sim h^H,$$

evaluated over the discretization sequence $N = 64, 128, 256, 512, 1024$.

A log-log regression of the empirical increment moments yields an estimated scaling exponent $\widehat{H} = 0.4765$ and $H_{\text{theoretical}} = 0.1100$.

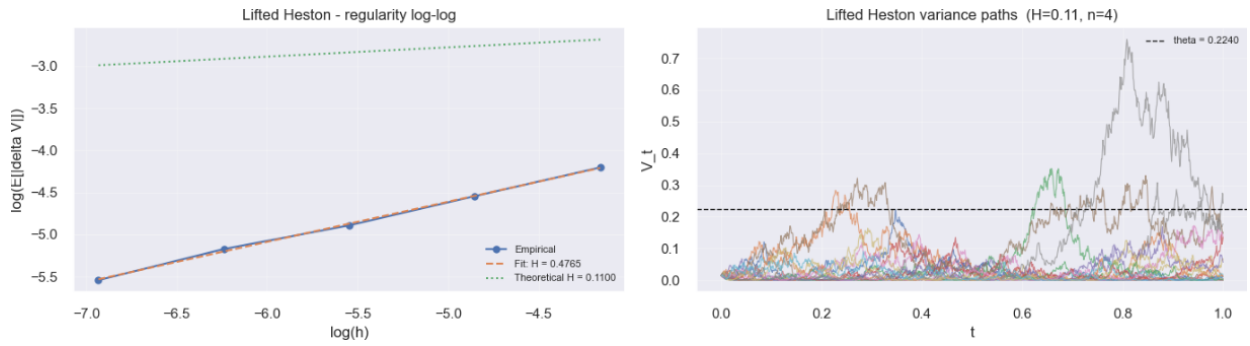

Figure 10: Lifted Heston regularity study: log-log scaling of variance increments and representative simulated variance paths.

The estimated exponent remains stable as the discretization becomes finer, indicating robust numerical scaling behavior. However, the observed regularity is much closer to the the standard Brownian case $H \approx 0.5$ than to the theoretical roughness level $H = 0.1100$. This happens because the finite number of factors makes the simulated paths smoother, even though the model still captures the main rough-volatility effects in calibration.

6 Joint SPX-VIX Calibration

To improve consistency between the SPX implied-volatility surface and the VIX market, we perform a joint calibration where the objective function combines the SPX fitting error with a VIX futures prices. To keep the two datasets balanced, the SPX surface is reduced to a smaller number of strikes per maturity, and the two loss components are equally weighted. For the classical Heston model, the joint calibration works well. The VIX RMSE improves significantly, decreasing from 63.01 to 27.26 vol points, while the SPX fit remains almost unchanged, moving only from 2.31 to 2.39 vol points. This means that the Heston model is able to slightly adjust its long-run variance level and improve the VIX fit without materially damaging the SPX calibration. The situation is different for Rough Heston. In this case, the joint calibration does not improve the results. The VIX RMSE worsens from 151.37 to 160.58 vol points, and the SPX RMSE also increases from 0.84 to 1.44 vol points. Therefore, the model is not able to fit both markets at the same time. The main reason is a structural limitation of the single-factor Rough Heston specification. The SPX-only calibration implies a long-run volatility level close to 29%, while the VIX futures curve converges to about 22%. Since the model has only one variance factor, the same long-run parameter must explain both the steep SPX skew and the lower VIX long-term level, creating a clear conflict. Overall, the joint calibration is effective for the classical Heston model, because it improves VIX consistency at almost no SPX cost. For Rough Heston, instead, the single-factor structure is too restrictive, and a richer multi-factor model would be needed to fit both.

7 Conclusions

This project compared the Heston, Rough Heston, and Lifted Heston models in their ability to reproduce both the SPX implied-volatility surface and VIX option smiles. The empirical analysis shows a clear trade-off between SPX fit accuracy, computational tractability, and consistency with the VIX market. The Heston model provides the most robust overall benchmark: although its SPX fit is less accurate than the rough specifications, it remains computationally efficient, numerically stable, and delivers the best VIX consistency among the SPX-only calibrations. The Rough Heston model strongly improves the SPX fit, especially in the overall smile shape and curvature across maturities, confirming the relevance of rough-volatility effects in equity index options. However, this improvement does not translate into a better VIX fit. Both the SPX-only and the joint calibration highlight a structural limitation of the single-factor specification: the same long-run variance level cannot simultaneously explain the steep SPX skew and the lower long-term level implied by VIX futures. The Lifted Heston model partially mitigates this issue. It preserves most of the SPX flexibility of the rough model while slightly improving the VIX fit and remaining easier to simulate thanks to its finite-dimensional structure. Still, the VIX mismatch remains significant. Overall, the results suggest that rough dynamics are essential for SPX smiles, but richer variance dynamics are needed for SPX-VIX consistency.

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